

Kohler Smart Facility & Sustainability Manager — Hospital Edition

By Mridul Choudhary

The Problem

High-footfall commercial facilities — hospitals, airports, universities — cannot manually monitor every fixture. Leaks, stuck valves, and hygiene lapses go undetected until they escalate into costly incidents or safety risks.

The Approach

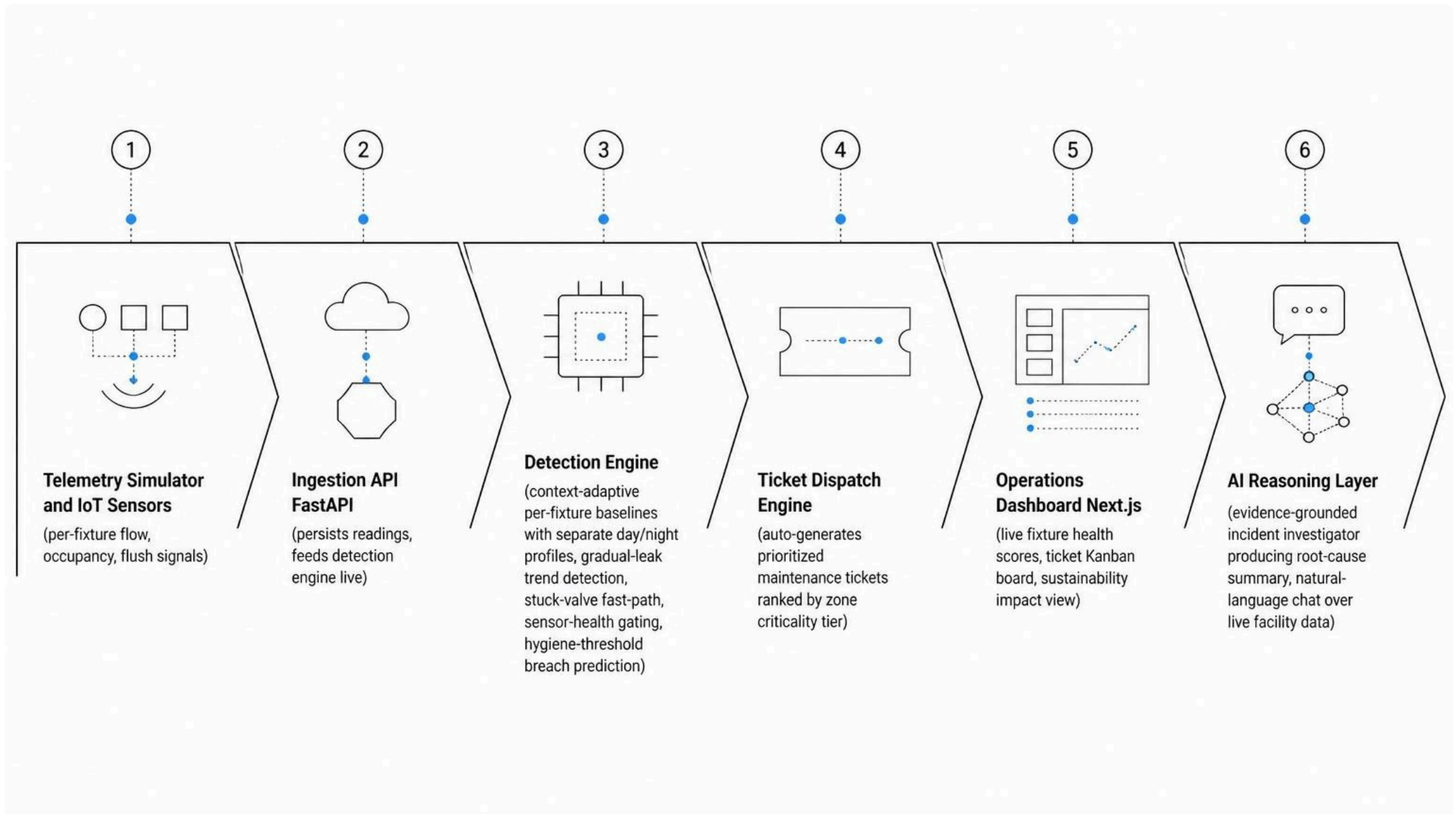
A real-time IoT monitoring and predictive dispatch platform, built incrementally across 8 phases — from telemetry ingestion through detection, prioritized ticket dispatch, and a live operations dashboard.

Validation

Built and tested against a 7-day simulated hospital dataset with labeled ground-truth anomalies and realistic "hard negative" scenarios — legitimate long hand-washes, traffic bursts — stress-testing false-positive resistance across four criticality tiers (ICU/OT = Tier 1 down to public lobbies = Tier 4).

System Architecture

A six-stage pipeline transforms raw fixture telemetry into prioritized maintenance actions and AI-grounded incident summaries — with sensor-health gating at every stage to prevent degraded hardware from triggering false alarms.



Each stage is independently testable; the detection engine and dispatch logic were validated against labeled anomaly scenarios before dashboard integration.

Tech Stack

Backend

Python · FastAPI (async) · SQLAlchemy 2.0 + Alembic migrations

Database

PostgreSQL 16

Frontend

Next.js 14 · TypeScript · Recharts

AI / LLM Layer

Groq-hosted LLM for incident summarization and chat-over-data, with strict evidence-grounding guardrails — the model only sees a deterministic evidence object assembled from real ticket and sensor data.

Simulator

Custom Python telemetry generator with injectable anomaly scenarios and ground-truth labeling for validation.

Testing

40+ automated unit and integration tests covering detection accuracy, false-positive resistance, and API contracts.

Why This Approach

Per-Fixture, Time-Aware Baselines

A night-shift ICU sink and a daytime lobby faucet are never compared against the same expectation.

Sensor-Health Gating

Degraded sensors route to maintenance, not a false leak ticket — eliminating the most damaging failure mode in facility monitoring.

Predictive Hygiene Forecasting

Acts *before* a threshold is crossed, not after — critical in hospital infection-control contexts.

Criticality-Tiered Dispatch

A Tier-1 ICU incident is never queued behind a low-priority lobby fixture.

Evidence-Grounded Root-Cause AI

Every claim in a generated summary traces to real sensor readings, avoiding the hallucination risk of ungrounded LLM outputs.

Transparent Sustainability Metrics

Water saved and cost avoided are computed with disclosed methodology — not presented as an unexplained number.

"A complete detect → diagnose → dispatch → quantify loop, validated end-to-end on realistic hospital-scale data." — **Mridul Choudhary**, KOHLER-MIT WPU AI Research Lab Program